Cochlear Implants

Learning Outcomes:

Understand the major components of a cochlear implant and function

Review the pre-operative procedures performed by the audiologist to determine candidacy for a cochlear implant

Identify members of the CI team and the importance of the SLP

Fundamental concept of a cochlear implant

- A cochlear implant is a device that electrically stimulates the auditory nerve of patients with **sensorineural** hearing loss to provide them with sound and speech information.
- Surgically implanted device that bypasses the peripheral auditory system to **directly electrically stimulate** the auditory nerve.
- It DOES NOT restore normal hearing
- Goal is to provide useful hearing and improved communication to adults and children with severe to profound hearing losses

Brief History

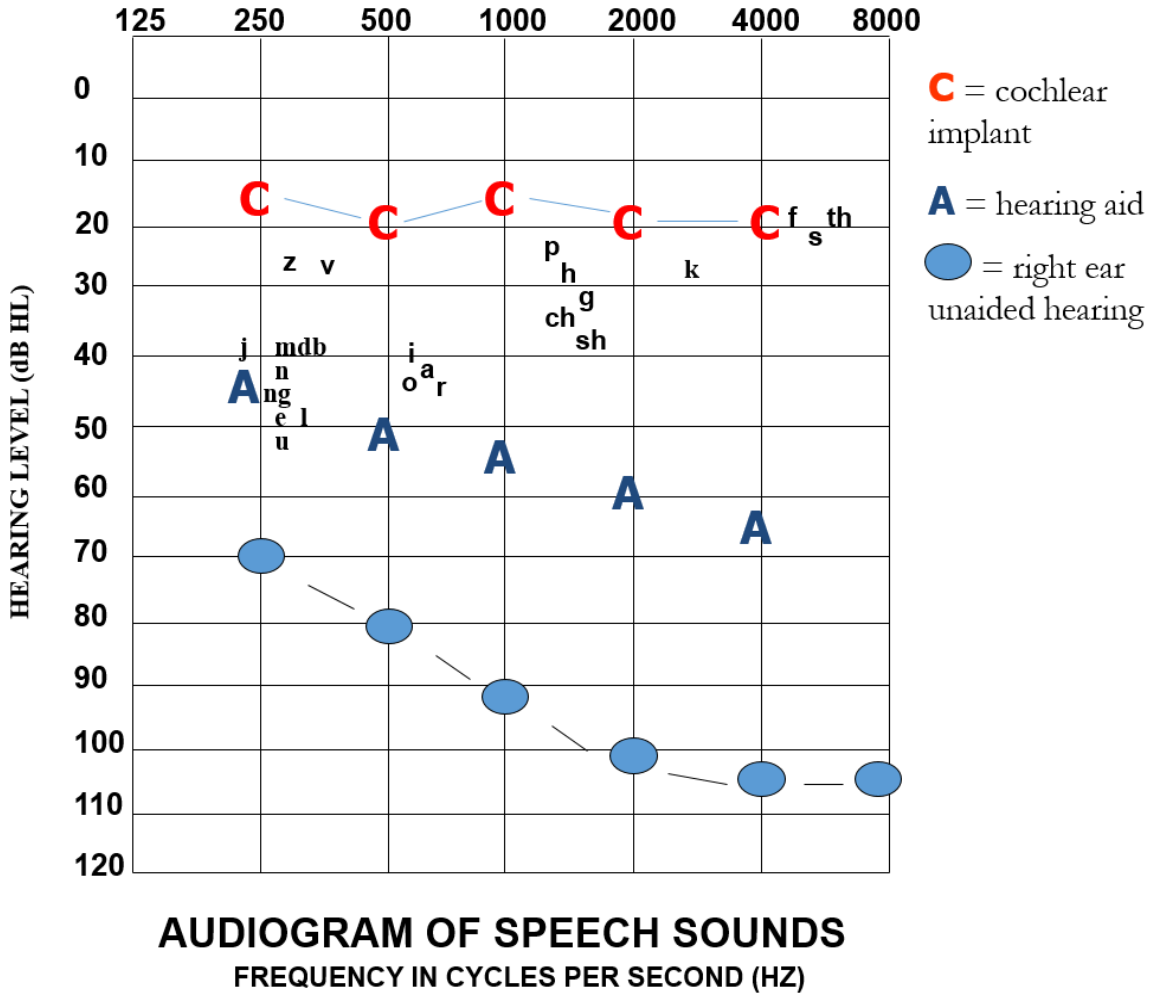
- Count Alessandro Volta in 1790 – experimented by putting electrodes in his ears
- Drs. Donald Eddington, William House, Robert Schindler, Robert White, and Graeme Clark, played pivotal roles in the evolution of cochlear implant technology.
- In 1972, one of the first wearable cochlear implants was implanted in Mr. Charles Graser at the House Ear Institute by Dr. William House. The House/3M device consisted of a single electrode implanted in the basal end of the cochlea with a ground electrode placed in the Eustachian tube.
- Multielectrode (or multichannel) cochlear implant devices first came into wide use in the 1980s.

Current Cochlear Implant Systems

- Three manufacturers - 2019, more than 736,900 cochlear implants worldwide had been registered from these companies
- Cochlear Americas
  - 1981 - development of a multichannel cochlear implant
  - This device was the Nucleus 22 cochlear implant (CI22M) and it secured approval from the USFDA in 1985.
- **MED-EL** Corporation
  - developed the world's first behind-the-ear (BTE) cochlear implant sound processor in 1991.
- Advanced Bionics
  - First cochlear implant system capable of simultaneous stimulation of multiple electrodes within the cochlea.

Why CI?

Up to **80%** of children who receive cochlear implants younger than **12** months demonstrated receptive vocabulary knowledge within the normal range by school entry. (Dettman SJ, et al. 2016)


125
250
500
1000
2000
4000
8000
0
10
20
30
40
50
60
70
80
90
100
110
120
HEARING LEVEL (dB HL)
C
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A
A
C = cochlear implant
A = hearing aid
= right ear unaided hearing
AUDIOGRAM OF SPEECH SOUNDS
FREQUENCY IN CYCLES PER SECOND (HZ)

Cochlear Implant Components

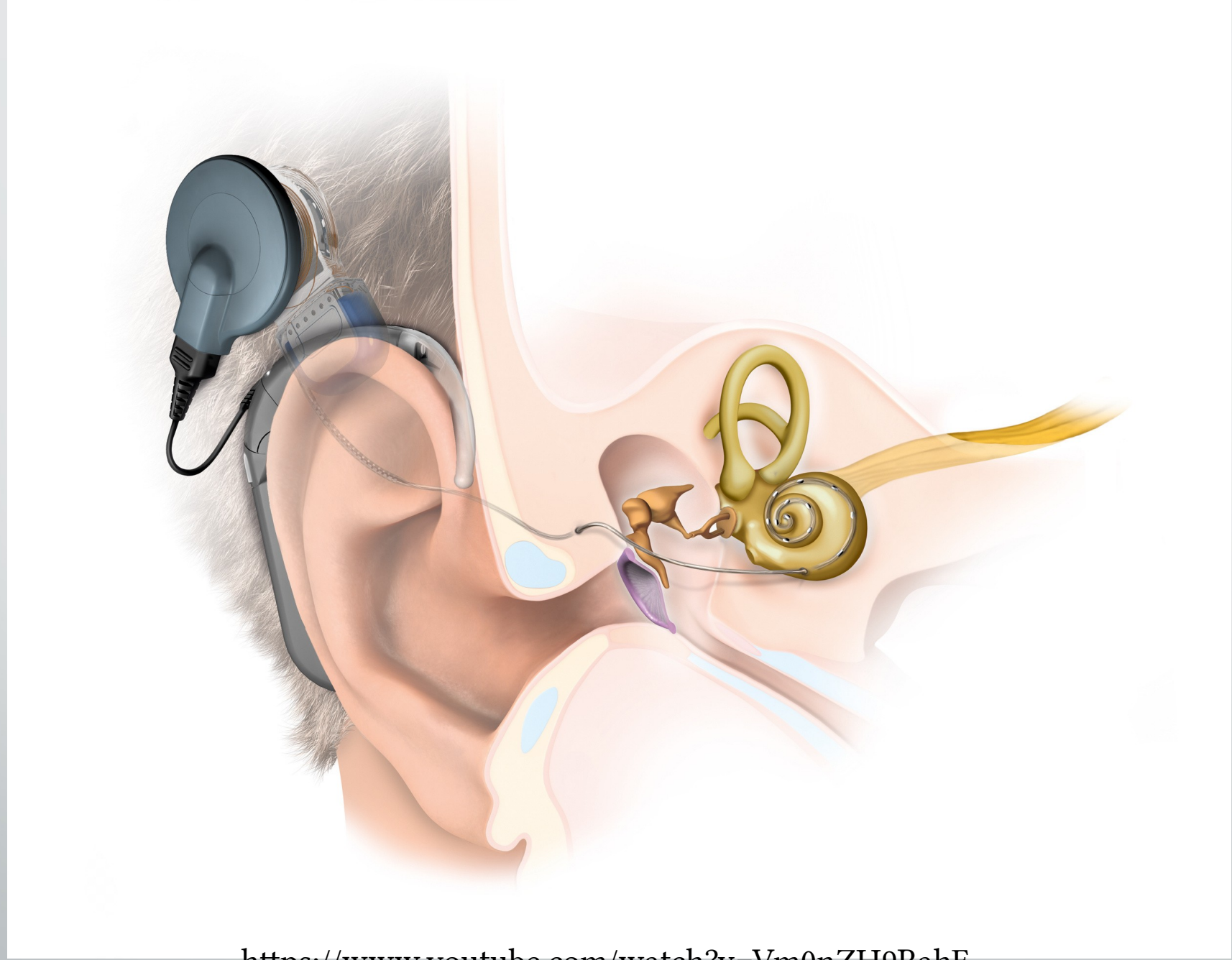
- External Components
  - Microphone
  - **Sound processor**
  - Transmitter
  - Battery
- Internal Components
  - **Receiver**
  - Electrode array

Internal and external CI components



Photo courtesy of Cochlear, Ltd.





https://www.youtube.com/watch?v=Vm0nZH9RahE

How does a Cochlear Implant Work?

- The microphone detects acoustic signals in the environment
- The sound processor converts the microphone's incoming signal into a **digital signal**
- The sound processor then relays the digital signal to the transmitter on the outside of the patient's scalp
- The transmitter in turn sends the digital signal to the **internal receiver** through the skin
- The receiver converts the digital signal into electrical pulses transmitted to the implanted electrode array
- Electrode array stimulates the auditory nerve fibers
- The auditory nerve sends the information to the brain

CI Candidacy

- Audiological Goals
  - Evaluate patient's hearing loss
  - Evaluate benefit from **hearing aids**
  - Work with other team members, especially the surgeon(s) and SLP(s) to determine if patient will benefit from a CI
  - Determine if patient meets FDA and/or insurer's Indications
  - Must have a **support system** in place and be motivated
  - Must have realistic expectations

Audiological Candidacy Criteria Adult (Table 3.1)

- Moderate to profound in the low Hz; severe to profound mid to high Hz
- Aided speech recognition scores
- Desire to be part of the hearing world
- No medical contraindications
- **High motivation**
- **Realistic expectations**

Audiological Candidacy Criteria young children (Table 3.1) less than 2 years

- **Bilateral Profound** SNHL bilaterally (Children 9-24 months)
- Lack of progress in the development of simple auditory skills when appropriately aided
- Therapy program that emphasizes the development of auditory skills
- **No medical contraindications**
- Highly motivated
- Parents have appropriate expectations

Audiological Candidacy criteria older children (Table 3.1) 2-18 years old

- Bilateral **severe to profound** SNHL bilaterally (2-18 years)**
- Lack of auditory and speech/language development with hearing aids
  - **≤30%** correct on open set word recognition tests in best aided condition
- Educational/therapy program that emphasizes the development of auditory skills
- No medical contraindications
- High motivation from patient and family
- Parent and child have appropriate expectations

Single sided deafness

- **5 years of age and older**
- Ear to be implanted
  - Severe to profound SNHL defined as: **Pure tone average ≥ 80 dB HL**
  - Aided CNC word score or developmentally appropriate word test < 30 dB HL
- Contralateral ear
  - Normal to near normal as defined as a pure tone average < 30 dB HL

When to consider a CI evaluation: infants and toddlers

- Does not react to **loud sounds**
- Does not seek out or detect where sound is coming from
- Has stopped babbling and experimenting with making sounds
- Still babbles but is **not progressing** to more understandable speech
- No reactions to voices, especially when being held.

When to consider a CI evaluation: School age

- Repeatedly misses or does not respond to **all Ling 6 sounds** with appropriately fit hearing technology
- Demonstrates increased dependency on remote microphone technology (FM/DM systems) or visual cues for access to speech in quiet environments
- Demonstrates slow auditory progress in comparison to **typical hearing peers**
- Easily frustrated or experiences communication breakdowns
- Exhausted at the end of the school day due to constant concentration just to understand speech

Audiological Evaluation

- Unaided testing
  - behavioral
  - **Electrophysiological (ABR/ASSR)**
- Aided testing(appropriately fit hearing aids)
  - **behavioral**
- Validation measure
  - IT-MAIS
  - CHILD
  - PEACH
  - ELF

Audiologic candidacy measure-pediatrics

Test measures depend on the age and language skills of the child



Evaluating very young children

- Children who receive a cochlear implant prior to **3** years of age, and especially prior to **18** months of age, make faster rates of progress with spoken language than children who receive their cochlear implant at older ages. Thus, it is important for us to provide implants early
- **ABR and behavioral testing** are used to confirm audiometric thresholds, and candidacy is typically based on these findings alone since speech perception testing is not possible.

Auditory Behavior/Functional Assessment

- Auditory Behavior/Functional assessments are used with children of all ages and necessary to see if the child meets FDA indications:
  - AB: failing to reach developmentally appropriate auditory milestones
  - Cochlear and MedEl: demonstrating lack of progress in simple auditory skill development.
- The information provided by the SLP, via observation and functional assessments, is also an essential part of determining if the child meets the indications cited above
- Audiological measures examine this as well:
  - IT-MAIS (Robbins, Renshaw, & Berry, 1991)
    - Birth-3 years, 10 parental questions that evaluate meaningful use of sound in everyday situations
  - MAIS (Robbins et al., 1991)
    - 3 years and up, 10 questions like the IT-MAIS
  - Little Ears (Kuehn-Inacken et al., 2003)
    - Birth and up, 35 parental questions that assess auditory development

Aided speech perception testing (Pediatrics)

Older Children: Open-Set Speech Recognition Measures

- Provide an unlimited number of response alternatives
- More difficult than closed-set tests (open-set)
- Chance score typically = 0%
- Should be presented recorded if possible
- Common measures = MLNT, LNT, Pediatric AzBio Sentences
- May be presented in quiet or in noise

Medical evaluation

- History
- Physical exam
- **genetics**
- Co existing conditions
- **Vaccines**
- Imaging
  - MRI
  - CT



Counseling

- Normal hearing
- CI hearing
- CI **timeline**
- Communication options
- Speech therapy
- **Support Groups**
- Realistic expectations
- CI manufacturers
- Immunizations
- Imaging
- Genetics
- Early Steps
- Parent Pupil Education Program
- Guide By Your Side
- Louisiana Hands and Voices

The CI Team

- Parents/family
- Audiologist
  - Responsible for assessing hearing function, determining device candidacy, and programming the CI **across the patient's lifespan**
- SLP
  - Help in developing the patient's speech and language skills, particularly for peds
- ENT
  - Plays a crucial role in performing the surgical implantation procedure and managing any medical aspects related to the implantation

Other CI team members could include:

- Psychologist
- Genetic counselor
- **Educator**
- Social worker
- **Patient navigator**
- Service coordinator

The CI Team

- CI Candidacy decisions require a team of professionals working together to make the best possible recommendation for a patient
- SLPs are valued members of CI teams; they provide important information that helps determine if a patient should receive an implant, if a patient is receiving appropriate benefit from an implant that helps guide future decisions regarding candidacy

Referrals for CI: SLP's are in the know

- Many CI programs rely on the expertise of the speech-language pathologist to help them determine if a patient is a candidate for a cochlear implant
- Similarly, many CI programs rely on the excellent referrals made by SLPs for candidacy evaluations. SLPs are often the first to know if a child is not making adequate progress in his/her spoken language skills and to notice if the child will/will not be able to keep up with his/her peers with typical hearing using his/her current hearing technology

Implantation

- Surgeon-
  - places electrode array
- Audiologist-
  - conducts telemetry (in real time verification)



Activation

- Initial activation **2-4** weeks after surgery
- Noise or voices like robots, Mickey Mouse, Donald Duck
- **Emotional reactions may vary**
- Emotional support
- Progressive maps put into programs
- Review all accessories in kit



Jonathan CI activation, 8 mths

- https://youtu.be/kHhvFlKyN-A
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Audiologic follow up

- Follow-up appointments
- Adjusting maps for comfortable loudness, sound quality, and **speech understanding**
- Sound field testing to ensure hearing thresholds and speech discrimination scores are improved
- 2 weeks, 1 month, 3 months, 6 months, annually

Successful outcomes

- **Intrinsic characteristics**
  - Gender
  - Cognitive ability
  - Family environment (SES)
  - **Age at onset of HL**
  - Pre-implant hearing
- **Implant characteristics**
  - Generation of Technology
  - Device Function
  - Duration of implant use
  - Age at implantation
- **Anatomy**
  - Boney structures, cochlea, and 8th nerve
- **Surgical procedure**
  - Size and anatomy of cochlea/cochleostomy site
  - **Depth of electrode insertion**
  - Electrode placement/type
- **Etiology of HL**
  - Spiral ganglion neurons (SGN) survival
  - Prior hearing aid use
- **Aural (re)habilitation**
  - Auditory Verbal Therapists (Certified LSLS)
  - Education Options

Hearing with Two ears

- Binaural hearing
  - Hearing with two ears
- **Bimodal hearing**
  - **Cochlear implant ✦ hearing aid**
- Bilateral hearing
  - Two cochlear implants

Timing Matters!

- Benefit varies from person to person
- Sooner is better than later
- Earlier cochlear implantation leads to:
  - Easier bonding with the device for the child.
  - Less time in therapy for child and parents.
  - Improved outcomes for speech, language, academics, and social skills.